STEVENS INSTITUTE of TECHNOLOGY
School of Business

FE 610

Problem Set 3

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For a Brownian motion $W(t)$, and times $0 < s < t < u$ answer the following:

1. Find $\mathbb{E}[W^2(t)W(s)|\mathcal{F}(s)]$
2. Find $\mathbb{E}[W^3(u) + W^2(t) + W(s)]$
3. Find $\mathbb{E}[W^4(u)] - \mathbb{E}[W^2(u)]^2$

Determine the following, assuming that $W(t)$ is a Brownian motion and that $0 < s < t$.

1. $\mathbb{E}[3W(t)]$
2. $\mathbb{E}[e^{3W(t)}]$
3. $\mathbb{E}[3W(t)|\mathcal{F}(s)]$
4. $\mathbb{E}[e^{3W(t)}|\mathcal{F}(s)]$

We know that Brownian motion is Markov. Therefore, we know that there exists some function g such that $\mathbb{E}[W^2(t)|\mathcal{F}(s)] = g(W(s))$. What is $g(x)$?

A stock that governed by **geometric Brownian motion** follows the structure

$$S(t) = S(0)e^{(\alpha - \frac{\sigma^2}{2})t + \sigma W(t)}$$

where $S(0)$, α, and σ are positive constants, is this process a martingale?

For a process $X(t) = 2W^2(t) - 5t^2$, is this process a martingale?

For a process, $S(t)$, that follows geometric Brownian motion, determine the following:

1. $\mathbb{E}[S^2(t)]$
2. $\mathbb{E}[S^3(t)]$
3. $\mathbb{E}[S^p(t)]$ where p is a positive constant.

For a Brownian motion $W(t)$, answer the following questions.

1. What is $\mathbb{P}(W(30) - W(5) > 10)$?
2. For a process $X(t) = 3W(t) + 7$, what is $[X, X](t)$?
3. For $s < t$, what is $\mathbb{E}[e^{2W(t)-t}|\mathcal{F}(s)]$?

For a Brownian motion $W(t)$ and times $s < t < u$, answer the following:

1. $\mathbb{E}[W(t)W(u)|\mathcal{F}(s)]$
2. $\mathbb{E}[(W(u) - W(t))^2 W(s) + W(u)W(t)W(s)]$